Zero-Day Anomaly Detection under HAI Plant-Version Shift

A three-seed, source-only evaluation of PCA, DeepSVDD, and a convolutional VAE with frozen and adaptive EVT decisions

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Abstract

This study evaluates zero-day anomaly detection on the HAI dataset. Its 59-tag public intersection is filtered using source-only variance to 41 nonconstant modeling sensors. The corpus contains 1,608,008 real telemetry rows and 102 documented attack events, including 52 in the held-out 23.05 domain. Models are trained only on normal 21.03 data and evaluated over three seeds (11, 23, 37). VAE achieved the highest mean 23.05 PR-AUC (0.232 ± 0.000). Frozen and adaptive PR-AUC are identical because both arms threshold the same continuous scores. For VAE, frozen event recall moved from 0.560 on 21.03 to 0.154 on 23.05; DSPOT changed cross-domain recall to 0.115 (gain -0.038; recovery ROM -0.095) and false alarms from 16.41 to 5.47 per day. DSPOT is interpreted strictly as decision-layer adaptation: it changes thresholds, not the learned representation.

Research Questions

- **RQ1 — Degradation:** How much do event detection, eTaPR F1, and point F1 degrade from 21.03 to 23.05 when the source threshold is frozen?
- **RQ2 — Recovery:** How much of that loss does source-calibrated DSPOT recover on 23.05, and at what observed false-alarm rate?
- **RQ3 — Ranking versus decisions:** Do threshold-independent ranking metrics support the same conclusions as thresholded event metrics?
- **RQ4 — Attribution:** How often do the top 3 anomalous sensors include a documented target retained in the source-nonconstant modeling schema?

Data and Experimental Protocol

Partition	Files	Rows	Duration (days at 1 Hz)	Attack events
21.03 normal training	3	921,603	10.667	0
21.03 test	5	402,005	4.653	50
23.05 test	2	284,400	3.292	52
All streams	10	1,608,008	18.611	102

The processed corpus comprises 1,608,008 one-hertz rows (18.611 stream-days). There are 50 attacks in 21.03 test and 52 in 23.05 test, for 102 total. The two plant versions share 59 public tags, but 18 have zero variance throughout the normal 21.03 training partitions. Their constant source encoding can dominate max-fused scores after plant-version shift. They are therefore removed using source-training variance alone, leaving 41 source-nonconstant modeling tags. Durations are computed from the actual per-file row counts in `data/processed/metadata.json`, rather than rounded summary values.

The three official 21.03 training files contain 921,603 rows, equivalent to 256.0 hours at 1 Hz. This measured duration differs from the 352 hours stated in HAI v4.0 documentation. All data accounting here follows the actual file row counts and per-file SHA-256 hashes recorded in metadata, not the copied documentation duration.

Normalization statistics are fitted only on chronological 21.03 training partitions. The last 15% of each normal source stream is reserved from model fitting and supplies normal calibration scores. No 23.05 labels, attack intervals, thresholds, or model updates are used during training or calibration. Evaluation replays each stream chronologically and separately for every model, seed, domain, and arm.

Methods

Shared windowing and model family

Each score is aligned to the causal end of a 60-second window; evaluation windows use a 10-second stride, while non-overlapping training windows use a 60-second stride. No future samples enter a window. DeepSVDD and the VAE use a matched three-layer, bias-free one-dimensional convolutional trunk and the same latent width. This architectural matching does not imply equal total parameter counts: the VAE additionally requires Gaussian heads and a decoder. DeepSVDD learns a latent hypersphere and scores squared distance to its fixed center. The standard VAE optimizes reconstruction plus KL divergence and scores reconstruction error using the mean latent at evaluation. The PCA baseline reconstructs flattened multivariate windows and derives per-sensor reconstruction errors. Per-sensor scores are fused by the configured maximum, so a system alarm requires only one strongly anomalous sensor.

Model	Seeds	Parameters	Epochs	Training seconds (mean ± SD)
PCA	11, 23, 37	n/a	n/a	0.88 ± 0.01
DeepSVDD	11, 23, 37	13904	40	14.79 ± 0.35
VAE	11, 23, 37	26352	40	30.30 ± 0.66

Parameter counts are read from each run's training metadata and show the actual SVDD/VAE capacity difference; PCA does not have a comparable neural parameter count. Training times are descriptive measurements, not controlled speed benchmarks.

EVT calibration and deployment arms

Both EVT arms are calibrated exclusively on normal 21.03 scores to an empirical target of 10 alarms/day. The initial threshold uses quantile 0.98, at least 20 exceedances, and 4000 initialization samples where available. The frozen arm deploys SPOT with its source-fitted terminal threshold. The adaptive arm deploys DSPOT with the same source calibration, then updates the decision threshold during chronological replay. DSPOT uses depth 360 and updates every 25 scored samples. **The EVT risk q is calibrated separately for frozen SPOT and adaptive DSPOT for every model and seed.** The observed calibration rate is an empirical selection criterion, not a guarantee of the shifted-domain false-alarm rate.

Model	Seed	Arm	q	Observed alarms/day	Target/day	Source threshold
PCA	11	Frozen SPOT	0.000184447	9.671	10.000	14.5997
PCA	11	Adaptive DSPOT	9.48091e-05	8.792	10.000	18.0041
PCA	23	Frozen SPOT	0.000184447	9.671	10.000	14.6003
PCA	23	Adaptive DSPOT	9.48091e-05	8.792	10.000	18.004
PCA	37	Frozen SPOT	0.000184447	9.671	10.000	14.602
PCA	37	Adaptive DSPOT	9.48091e-05	8.792	10.000	18.0021
DeepSVDD	11	Frozen SPOT	0.00135811	10.551	10.000	1.16553
DeepSVDD	11	Adaptive DSPOT	0.00264215	15.826	10.000	0.914898
DeepSVDD	23	Frozen SPOT	0.00115741	15.826	10.000	1.36922
DeepSVDD	23	Adaptive DSPOT	0.00135811	10.551	10.000	1.28297
DeepSVDD	37	Frozen SPOT	0.00115741	14.067	10.000	1.81606
DeepSVDD	37	Adaptive DSPOT	0.00115741	12.309	10.000	1.40333
VAE	11	Frozen SPOT	4.87336e-05	10.551	10.000	36.2371
VAE	11	Adaptive DSPOT	9.48091e-05	10.551	10.000	27.1587
VAE	23	Frozen SPOT	4.87336e-05	10.551	10.000	36.3569
VAE	23	Adaptive DSPOT	9.48091e-05	10.551	10.000	27.3739
VAE	37	Frozen SPOT	4.87336e-05	10.551	10.000	36.3098
VAE	37	Adaptive DSPOT	9.48091e-05	10.551	10.000	26.8645

Metrics and attribution

PR-AUC and ROC-AUC are computed from continuous scores and are threshold-independent. Point precision, recall, and F1 use raw thresholded points with **no point adjustment**. eTaPR uses theta_p = 0.5 and theta_r = 0.1; event detection recall counts an attack as detected when at least one alarm occurs within its documented interval. Detection delay is measured from event start to the first in-event alarm and is summarized only over detected attacks. False alarms are alarms on normal points, normalized by the actual evaluated stream-days.

eTaPR is the enhanced time-series-aware metric published at SAC 2022 [7], not the CIKM 2019 TaPR metric [6]. TaPR and eTaPR are distinct; this report does not relabel either citation or use point adjustment, whose optimistic bias is discussed by Kim et al. [8].

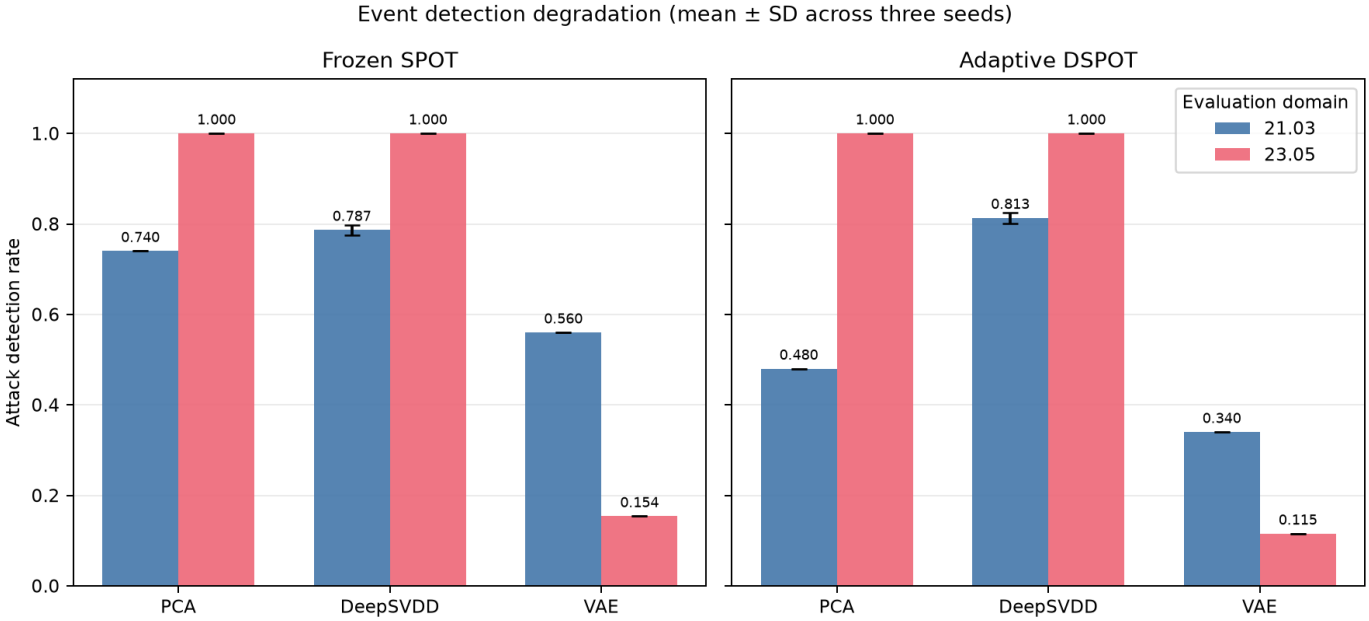
Attribution averages per-sensor scores over each attack interval and ranks sensors. Hit@3 is true when any retained documented target appears in the top 3. A retained target is the intersection of attack documentation with the 41-tag source-nonconstant modeling schema; target coverage is computed against this same schema. **If that intersection is empty, the attack is excluded from the attribution-hit denominator and is not counted as an attribution failure.** Target coverage is reported separately so this eligibility restriction remains visible.

Results

Three-seed aggregate results

Entries are mean ± sample standard deviation across the three seeds. Event recall is the documented-attack detection rate; FA/day is the false-alarm rate.

Model	Domain	Arm	Event recall	eTa F1	Point F1	PR-AUC	FA/day	Hit@3
PCA	21.03	Frozen	0.740 ± 0.000	0.533 ± 0.000	0.548 ± 0.000	0.502 ± 0.000	30.11 ± 0.00	0.229 ± 0.000
PCA	21.03	Adaptive	0.480 ± 0.000	0.386 ± 0.000	0.370 ± 0.000	0.502 ± 0.000	18.93 ± 0.00	0.229 ± 0.000
PCA	23.05	Frozen	1.000 ± 0.000	0.000 ± 0.000	0.076 ± 0.000	0.110 ± 0.000	8296.59 ± 0.00	0.033 ± 0.000
PCA	23.05	Adaptive	1.000 ± 0.000	0.000 ± 0.000	0.076 ± 0.000	0.110 ± 0.000	8296.59 ± 0.00	0.033 ± 0.000
DeepSVDD	21.03	Frozen	0.787 ± 0.012	0.533 ± 0.036	0.614 ± 0.011	0.486 ± 0.002	73.98 ± 12.16	0.152 ± 0.072
DeepSVDD	21.03	Adaptive	0.813 ± 0.012	0.528 ± 0.006	0.607 ± 0.009	0.486 ± 0.002	85.74 ± 11.71	0.152 ± 0.072
DeepSVDD	23.05	Frozen	1.000 ± 0.000	0.000 ± 0.000	0.076 ± 0.000	0.092 ± 0.003	8296.59 ± 0.00	0.100 ± 0.000
DeepSVDD	23.05	Adaptive	1.000 ± 0.000	0.000 ± 0.000	0.076 ± 0.000	0.092 ± 0.003	8296.59 ± 0.00	0.100 ± 0.000
VAE	21.03	Frozen	0.560 ± 0.000	0.473 ± 0.000	0.313 ± 0.001	0.462 ± 0.001	18.28 ± 0.00	0.257 ± 0.000
VAE	21.03	Adaptive	0.340 ± 0.000	0.319 ± 0.001	0.218 ± 0.003	0.462 ± 0.001	13.91 ± 0.50	0.257 ± 0.000
VAE	23.05	Frozen	0.154 ± 0.000	0.103 ± 0.000	0.216 ± 0.000	0.232 ± 0.000	16.41 ± 0.00	0.200 ± 0.000
VAE	23.05	Adaptive	0.115 ± 0.000	0.095 ± 0.000	0.089 ± 0.002	0.232 ± 0.000	5.47 ± 0.00	0.200 ± 0.000



Event detection rates on 21.03 and 23.05 for both deployment arms.

Complete run-level results

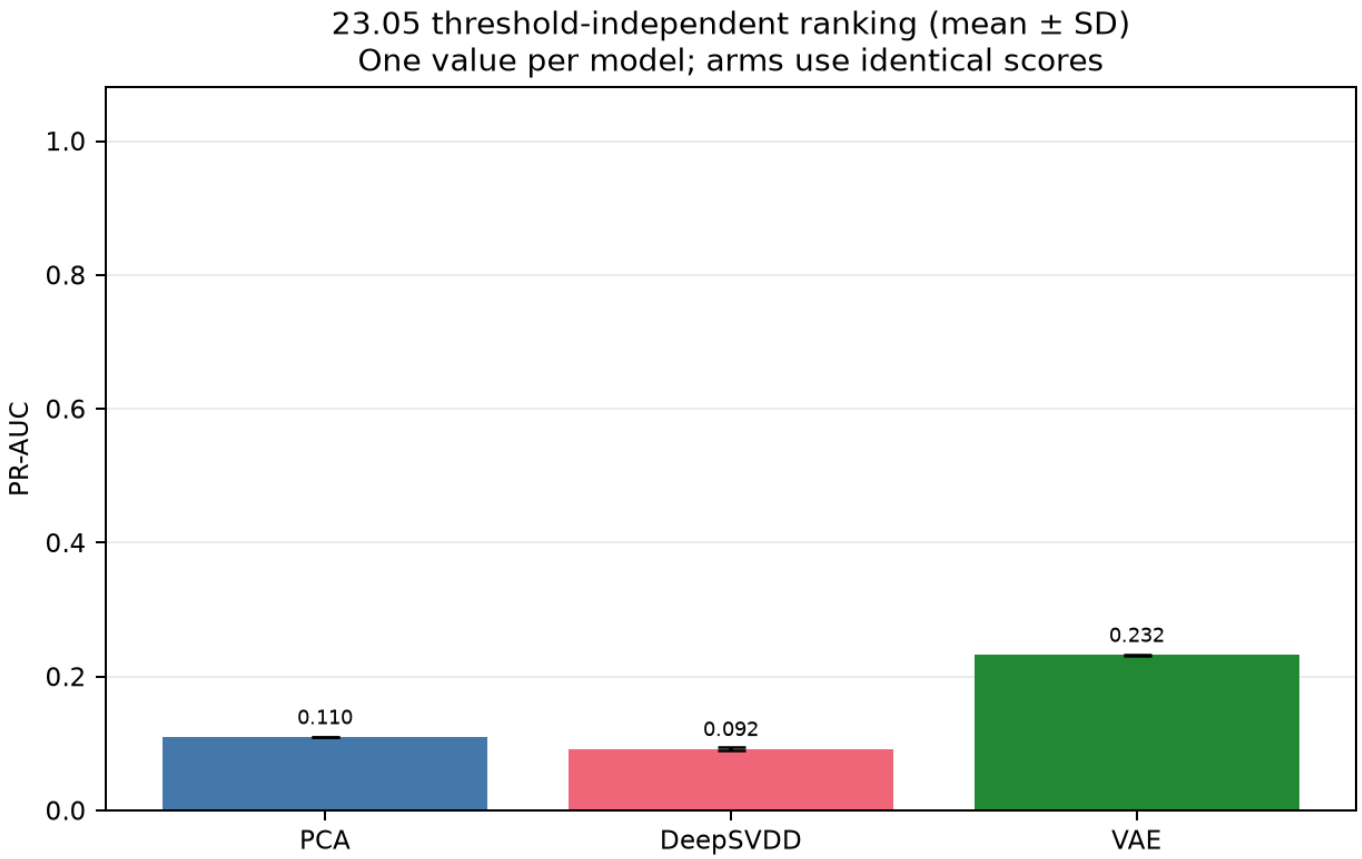
The following tables include every model, seed, domain, and arm. No run is hidden behind the aggregate summary.

Event detection and eTaPR

Model	Seed	Domain	Arm	Detected	Event recall	eTa P	eTa R	eTa F1	eTa det. R
PCA	11	21.03	Frozen	37/50	0.740	0.671	0.442	0.533	0.560
PCA	11	21.03	Adaptive	24/50	0.480	0.671	0.270	0.386	0.360
PCA	11	23.05	Frozen	52/52	1.000	0.000	0.000	0.000	0.000
PCA	11	23.05	Adaptive	52/52	1.000	0.000	0.000	0.000	0.000
PCA	23	21.03	Frozen	37/50	0.740	0.671	0.442	0.533	0.560
PCA	23	21.03	Adaptive	24/50	0.480	0.671	0.270	0.386	0.360
PCA	23	23.05	Frozen	52/52	1.000	0.000	0.000	0.000	0.000
PCA	23	23.05	Adaptive	52/52	1.000	0.000	0.000	0.000	0.000
PCA	37	21.03	Frozen	37/50	0.740	0.671	0.442	0.533	0.560
PCA	37	21.03	Adaptive	24/50	0.480	0.671	0.270	0.386	0.360
PCA	37	23.05	Frozen	52/52	1.000	0.000	0.000	0.000	0.000
PCA	37	23.05	Adaptive	52/52	1.000	0.000	0.000	0.000	0.000
DeepSVDD	11	21.03	Frozen	40/50	0.800	0.500	0.561	0.529	0.640
DeepSVDD	11	21.03	Adaptive	41/50	0.820	0.463	0.607	0.525	0.700
DeepSVDD	11	23.05	Frozen	52/52	1.000	0.000	0.000	0.000	0.000
DeepSVDD	11	23.05	Adaptive	52/52	1.000	0.000	0.000	0.000	0.000
DeepSVDD	23	21.03	Frozen	39/50	0.780	0.462	0.545	0.500	0.600
DeepSVDD	23	21.03	Adaptive	40/50	0.800	0.484	0.570	0.524	0.640
DeepSVDD	23	23.05	Frozen	52/52	1.000	0.000	0.000	0.000	0.000
DeepSVDD	23	23.05	Adaptive	52/52	1.000	0.000	0.000	0.000	0.000
DeepSVDD	37	21.03	Frozen	39/50	0.780	0.581	0.561	0.571	0.640
DeepSVDD	37	21.03	Adaptive	41/50	0.820	0.509	0.562	0.534	0.640
DeepSVDD	37	23.05	Frozen	52/52	1.000	0.000	0.000	0.000	0.000
DeepSVDD	37	23.05	Adaptive	52/52	1.000	0.000	0.000	0.000	0.000
VAE	11	21.03	Frozen	28/50	0.560	0.688	0.361	0.473	0.520
VAE	11	21.03	Adaptive	17/50	0.340	0.612	0.215	0.318	0.300
VAE	11	23.05	Frozen	8/52	0.154	0.583	0.057	0.103	0.077
VAE	11	23.05	Adaptive	6/52	0.115	0.620	0.051	0.094	0.077
VAE	23	21.03	Frozen	28/50	0.560	0.687	0.361	0.473	0.520
VAE	23	21.03	Adaptive	17/50	0.340	0.609	0.216	0.319	0.300
VAE	23	23.05	Frozen	8/52	0.154	0.583	0.057	0.103	0.077
VAE	23	23.05	Adaptive	6/52	0.115	0.643	0.051	0.095	0.077
VAE	37	21.03	Frozen	28/50	0.560	0.688	0.361	0.473	0.520
VAE	37	21.03	Adaptive	17/50	0.340	0.609	0.216	0.319	0.300
VAE	37	23.05	Frozen	8/52	0.154	0.583	0.057	0.103	0.077
VAE	37	23.05	Adaptive	6/52	0.115	0.643	0.051	0.095	0.077

Point-wise and threshold-independent metrics

Model	Seed	Domain	Arm	Point P	Point R	Point F1	PR-AUC	ROC-AUC
PCA	11	21.03	Frozen	0.735	0.436	0.548	0.502	0.867
PCA	11	21.03	Adaptive	0.716	0.250	0.370	0.502	0.867
PCA	11	23.05	Frozen	0.040	1.000	0.076	0.110	0.499
PCA	11	23.05	Adaptive	0.040	1.000	0.076	0.110	0.499
PCA	23	21.03	Frozen	0.735	0.436	0.548	0.502	0.867
PCA	23	21.03	Adaptive	0.716	0.250	0.370	0.502	0.867
PCA	23	23.05	Frozen	0.040	1.000	0.076	0.110	0.499
PCA	23	23.05	Adaptive	0.040	1.000	0.076	0.110	0.499
PCA	37	21.03	Frozen	0.735	0.436	0.548	0.502	0.867
PCA	37	21.03	Adaptive	0.716	0.250	0.370	0.502	0.867
PCA	37	23.05	Frozen	0.040	1.000	0.076	0.110	0.499
PCA	37	23.05	Adaptive	0.040	1.000	0.076	0.110	0.499
DeepSVDD	11	21.03	Frozen	0.597	0.619	0.608	0.485	0.840
DeepSVDD	11	21.03	Adaptive	0.556	0.645	0.597	0.485	0.840
DeepSVDD	11	23.05	Frozen	0.040	1.000	0.076	0.089	0.424
DeepSVDD	11	23.05	Adaptive	0.040	1.000	0.076	0.089	0.424
DeepSVDD	23	21.03	Frozen	0.591	0.622	0.606	0.488	0.853
DeepSVDD	23	21.03	Adaptive	0.592	0.632	0.612	0.488	0.853
DeepSVDD	23	23.05	Frozen	0.040	1.000	0.076	0.094	0.447
DeepSVDD	23	23.05	Adaptive	0.040	1.000	0.076	0.094	0.447
DeepSVDD	37	21.03	Frozen	0.656	0.600	0.627	0.484	0.844
DeepSVDD	37	21.03	Adaptive	0.610	0.618	0.614	0.484	0.844
DeepSVDD	37	23.05	Frozen	0.040	1.000	0.076	0.092	0.428
DeepSVDD	37	23.05	Adaptive	0.040	1.000	0.076	0.092	0.428
VAE	11	21.03	Frozen	0.680	0.204	0.313	0.461	0.871
VAE	11	21.03	Adaptive	0.648	0.128	0.214	0.461	0.871
VAE	11	23.05	Frozen	0.726	0.127	0.216	0.232	0.628
VAE	11	23.05	Adaptive	0.743	0.046	0.087	0.232	0.628
VAE	23	21.03	Frozen	0.679	0.202	0.312	0.462	0.871
VAE	23	21.03	Adaptive	0.641	0.133	0.220	0.462	0.871
VAE	23	23.05	Frozen	0.726	0.127	0.216	0.233	0.644
VAE	23	23.05	Adaptive	0.750	0.048	0.090	0.233	0.644
VAE	37	21.03	Frozen	0.680	0.204	0.313	0.462	0.871
VAE	37	21.03	Adaptive	0.641	0.133	0.220	0.462	0.871
VAE	37	23.05	Frozen	0.726	0.127	0.216	0.232	0.645
VAE	37	23.05	Adaptive	0.750	0.048	0.090	0.232	0.645

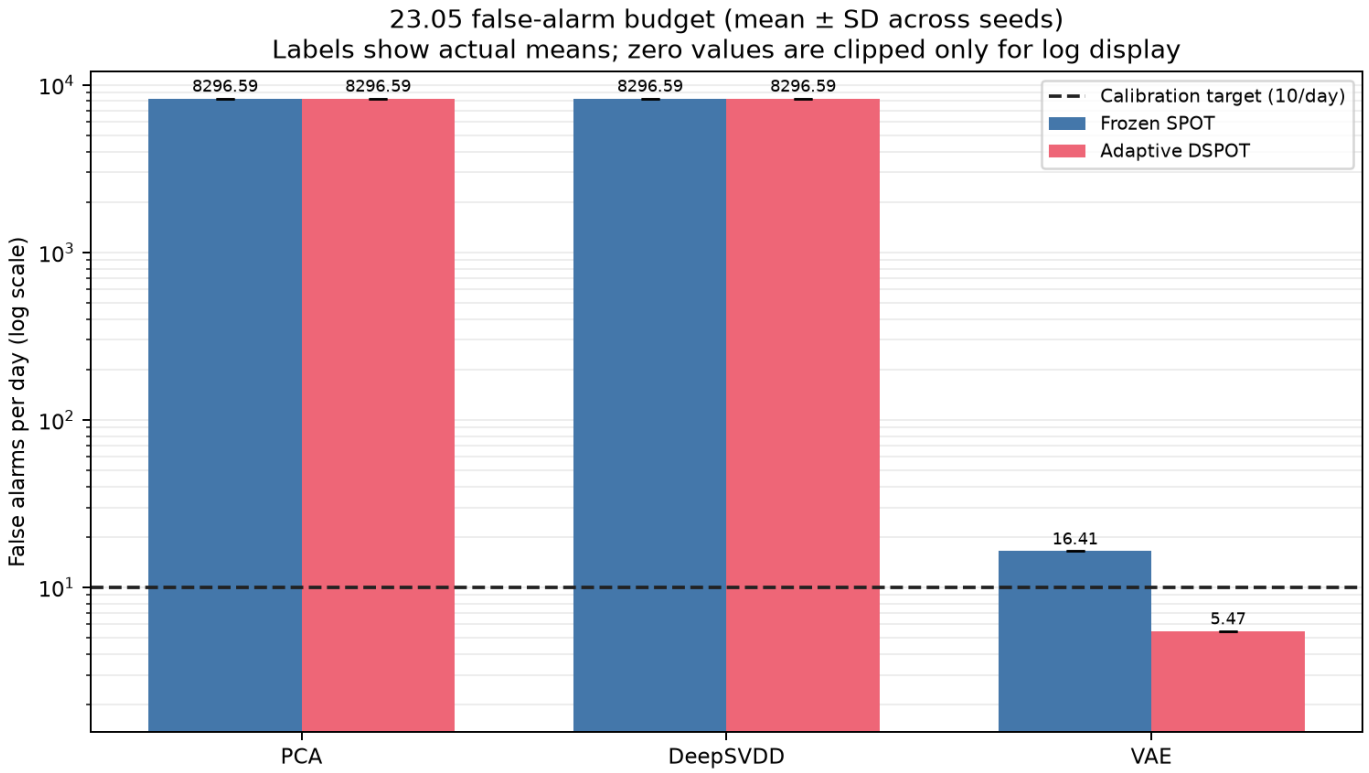


Cross-domain PR-AUC without duplicate arm bars.

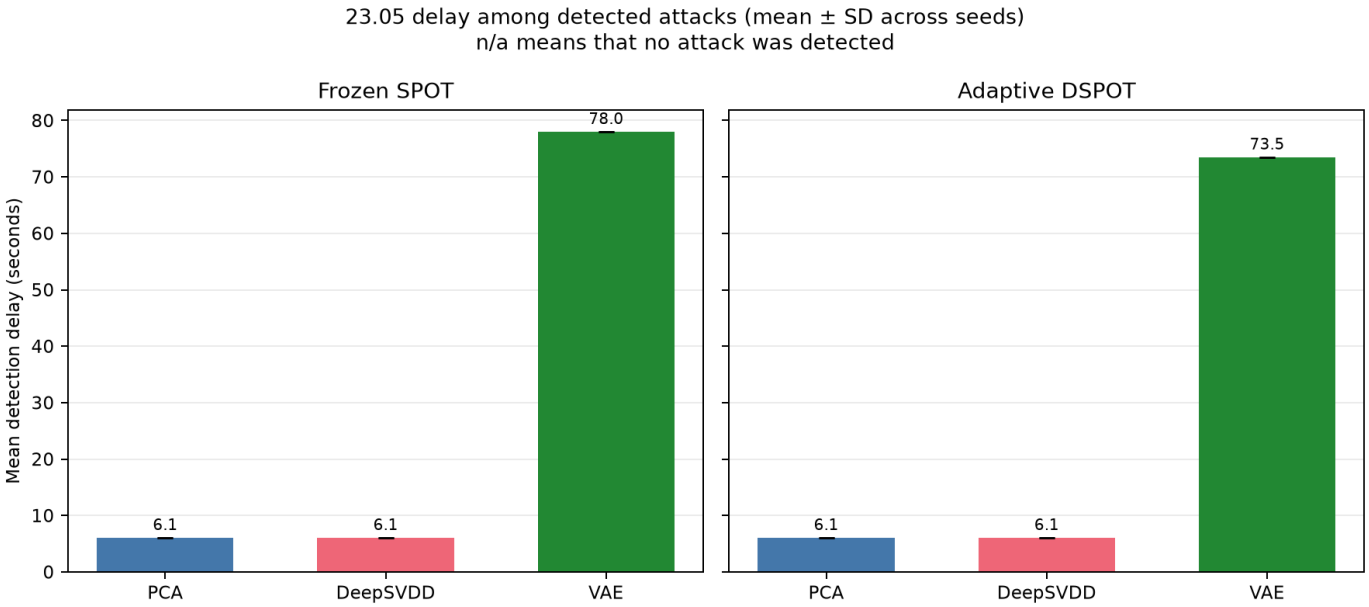
The PR-AUC figure intentionally has one bar per model rather than duplicate frozen/adaptive bars. Both arms consume exactly the same continuous anomaly scores; only their thresholds and resulting decisions differ. Consequently their PR-AUC and ROC-AUC values are identical for a given model, seed, and domain.

Alarm budget, delay, and attribution

Model	Seed	Domain	Arm	Alarms	False alarms	FA/day	Days	Mean delay (s)	Median delay (s)	Target coverage	Hit@3
PCA	11	21.03	Frozen	528	140	30.11	4.650	49.4	39.0	0.700	0.229
PCA	11	21.03	Adaptive	310	88	18.93	4.650	55.8	46.0	0.700	0.229
PCA	11	23.05	Frozen	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.033
PCA	11	23.05	Adaptive	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.033
PCA	23	21.03	Frozen	528	140	30.11	4.650	49.4	39.0	0.700	0.229
PCA	23	21.03	Adaptive	310	88	18.93	4.650	55.8	46.0	0.700	0.229
PCA	23	23.05	Frozen	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.033
PCA	23	23.05	Adaptive	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.033
PCA	37	21.03	Frozen	528	140	30.11	4.650	49.4	39.0	0.700	0.229
PCA	37	21.03	Adaptive	310	88	18.93	4.650	55.8	46.0	0.700	0.229
PCA	37	23.05	Frozen	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.033
PCA	37	23.05	Adaptive	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.033
DeepSVDD	11	21.03	Frozen	921	371	79.79	4.650	39.6	28.0	0.700	0.143
DeepSVDD	11	21.03	Adaptive	1031	458	98.50	4.650	36.9	27.0	0.700	0.143
DeepSVDD	11	23.05	Frozen	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.100
DeepSVDD	11	23.05	Adaptive	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.100
DeepSVDD	23	21.03	Frozen	935	382	82.15	4.650	37.1	28.0	0.700	0.229
DeepSVDD	23	21.03	Adaptive	949	387	83.23	4.650	36.9	27.0	0.700	0.229
DeepSVDD	23	23.05	Frozen	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.100
DeepSVDD	23	23.05	Adaptive	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.100
DeepSVDD	37	21.03	Frozen	812	279	60.00	4.650	38.6	30.0	0.700	0.086
DeepSVDD	37	21.03	Adaptive	900	351	75.49	4.650	42.5	33.0	0.700	0.086
DeepSVDD	37	23.05	Frozen	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.100
DeepSVDD	37	23.05	Adaptive	28430	27300	8296.59	3.291	6.1	6.5	0.577	0.100
VAE	11	21.03	Frozen	266	85	18.28	4.650	62.8	46.5	0.700	0.257
VAE	11	21.03	Adaptive	176	62	13.33	4.650	63.4	49.0	0.700	0.257
VAE	11	23.05	Frozen	197	54	16.41	3.291	78.0	45.0	0.577	0.200
VAE	11	23.05	Adaptive	70	18	5.47	3.291	73.5	44.5	0.577	0.200
VAE	23	21.03	Frozen	265	85	18.28	4.650	63.1	46.5	0.700	0.257
VAE	23	21.03	Adaptive	184	66	14.19	4.650	62.2	49.0	0.700	0.257
VAE	23	23.05	Frozen	197	54	16.41	3.291	78.0	45.0	0.577	0.200
VAE	23	23.05	Adaptive	72	18	5.47	3.291	73.5	44.5	0.577	0.200
VAE	37	21.03	Frozen	266	85	18.28	4.650	62.8	46.5	0.700	0.257
VAE	37	21.03	Adaptive	184	66	14.19	4.650	62.8	49.0	0.700	0.257
VAE	37	23.05	Frozen	197	54	16.41	3.291	78.0	45.0	0.577	0.200
VAE	37	23.05	Adaptive	72	18	5.47	3.291	73.5	44.5	0.577	0.200



Cross-domain false-alarm rates on a logarithmic axis with the calibration target.



Cross-domain detection delay among detected attacks.

Among conditions with at least one detected attack, PCA / frozen and PCA / adaptive and DeepSVDD / frozen and DeepSVDD / adaptive had the lowest across-seed mean of per-run mean delays (6.1 ± 0.0 s). Delay is conditional on detection and therefore does not penalize missed attacks.

Degradation and DSPOT Recovery

For each model, seed, and metric, degradation is `loss = metric_21_frozen - metric_23_frozen`, and DSPOT gain is `gain = metric_23_adaptive - metric_23_frozen`. Positive loss denotes degradation and positive gain denotes improvement. Per-seed recovery is `gain / loss` **only when loss > 0**; otherwise it is undefined and excluded from the mean-of-ratios. The final column instead reports the ratio of mean gain to mean loss, which uses all seeds and is defined only when mean loss > 0. These two recovery estimators answer different questions and need not agree.

Model	Metric	21.03 F	23.05 F	23.05 A	Loss	Gain	Mean recovery	Gain/loss ROM
PCA	Event recall	0.740 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	-0.260 ± 0.000	0.000 ± 0.000	n/a ± n/a	n/a
PCA	eTa F1	0.533 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.533 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000
PCA	Point F1	0.548 ± 0.000	0.076 ± 0.000	0.076 ± 0.000	0.471 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000
DeepSVDD	Event recall	0.787 ± 0.012	1.000 ± 0.000	1.000 ± 0.000	-0.213 ± 0.012	0.000 ± 0.000	n/a ± n/a	n/a
DeepSVDD	eTa F1	0.533 ± 0.036	0.000 ± 0.000	0.000 ± 0.000	0.533 ± 0.036	0.000 ± 0.000	0.000 ± 0.000	0.000
DeepSVDD	Point F1	0.614 ± 0.011	0.076 ± 0.000	0.076 ± 0.000	0.537 ± 0.011	0.000 ± 0.000	0.000 ± 0.000	0.000
VAE	Event recall	0.560 ± 0.000	0.154 ± 0.000	0.115 ± 0.000	0.406 ± 0.000	-0.038 ± 0.000	-0.095 ± 0.000	-0.095
VAE	eTa F1	0.473 ± 0.000	0.103 ± 0.000	0.095 ± 0.000	0.370 ± 0.000	-0.008 ± 0.000	-0.023 ± 0.001	-0.023
VAE	Point F1	0.313 ± 0.001	0.216 ± 0.000	0.089 ± 0.002	0.097 ± 0.001	-0.127 ± 0.002	-1.301 ± 0.016	-1.301

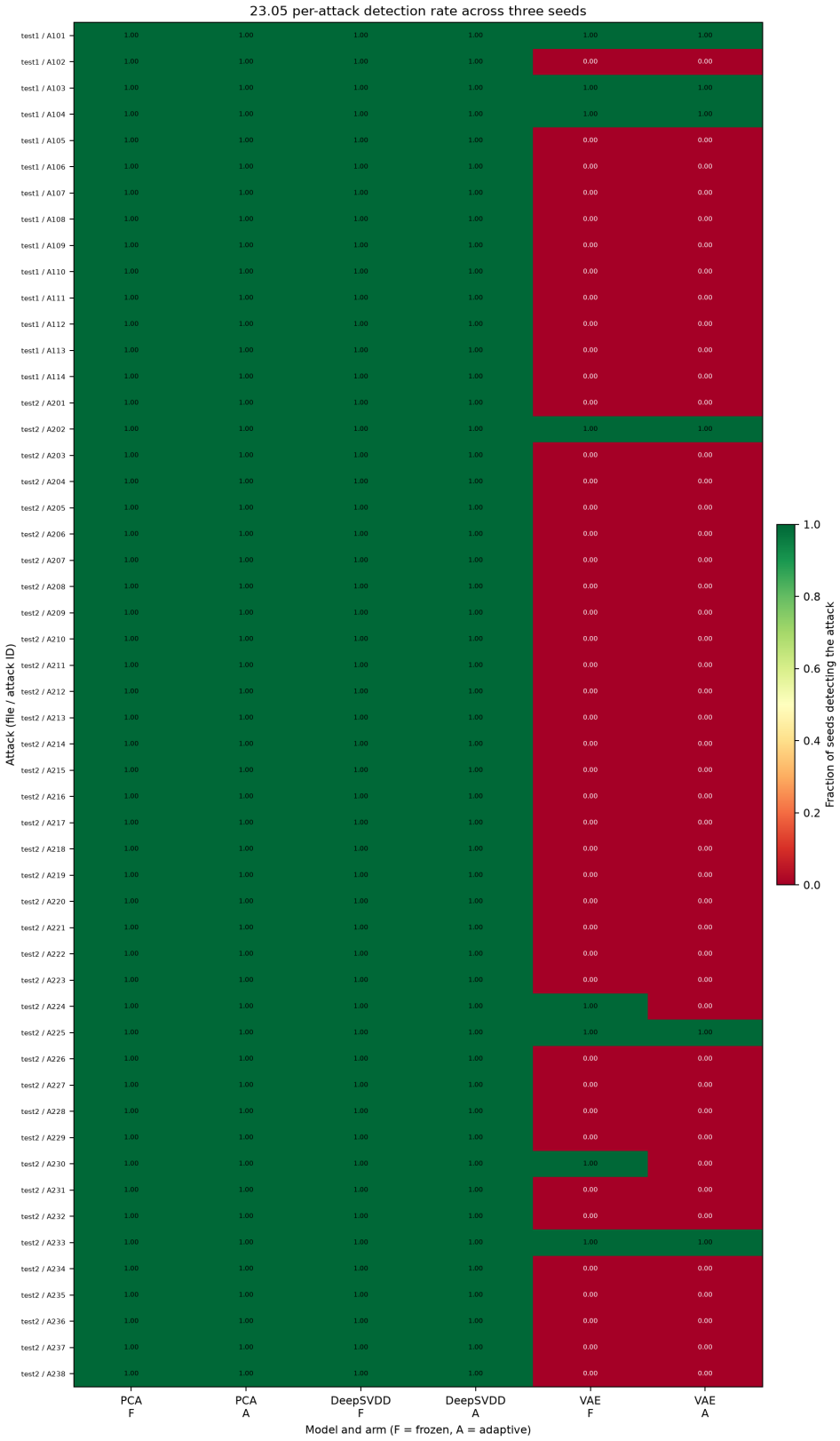
In the table, F and A denote frozen and adaptive arms; ROM denotes ratio of means. `n/a` marks an undefined ratio, not zero recovery. Per-seed values are provided in `degradation_recovery.csv`; aggregate values and valid-seed counts are in `recovery_summary.csv`.

Per-Attack Detection Failures

Across the 52 23.05 attacks, 0 were missed in every model/seed/arm run and 46 had at least one miss but at least one detection. The table lists every attack with any failed run. Cells give detected seeds / total seeds; attacks not listed were detected in every run.

Attack	File	Duration (s)	Missed runs	PCA F	PCA A	DeepSVDD F	DeepSVDD A	VAE F	VAE A
A102	test1	199	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A105	test1	162	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A106	test1	198	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A107	test1	605	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A108	test1	97	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A109	test1	131	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A110	test1	56	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A111	test1	132	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A112	test1	79	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A113	test1	134	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A114	test1	628	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A201	test2	133	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A203	test2	69	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A204	test2	123	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A205	test2	86	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A206	test2	197	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A207	test2	615	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A208	test2	134	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A209	test2	86	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A210	test2	89	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A211	test2	205	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A212	test2	128	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A213	test2	540	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A214	test2	62	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A215	test2	148	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A216	test2	96	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A217	test2	506	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A218	test2	215	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A219	test2	132	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A220	test2	132	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A221	test2	83	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A222	test2	212	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A223	test2	80	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A224	test2	108	3/18	3/3	3/3	3/3	3/3	3/3	0/3

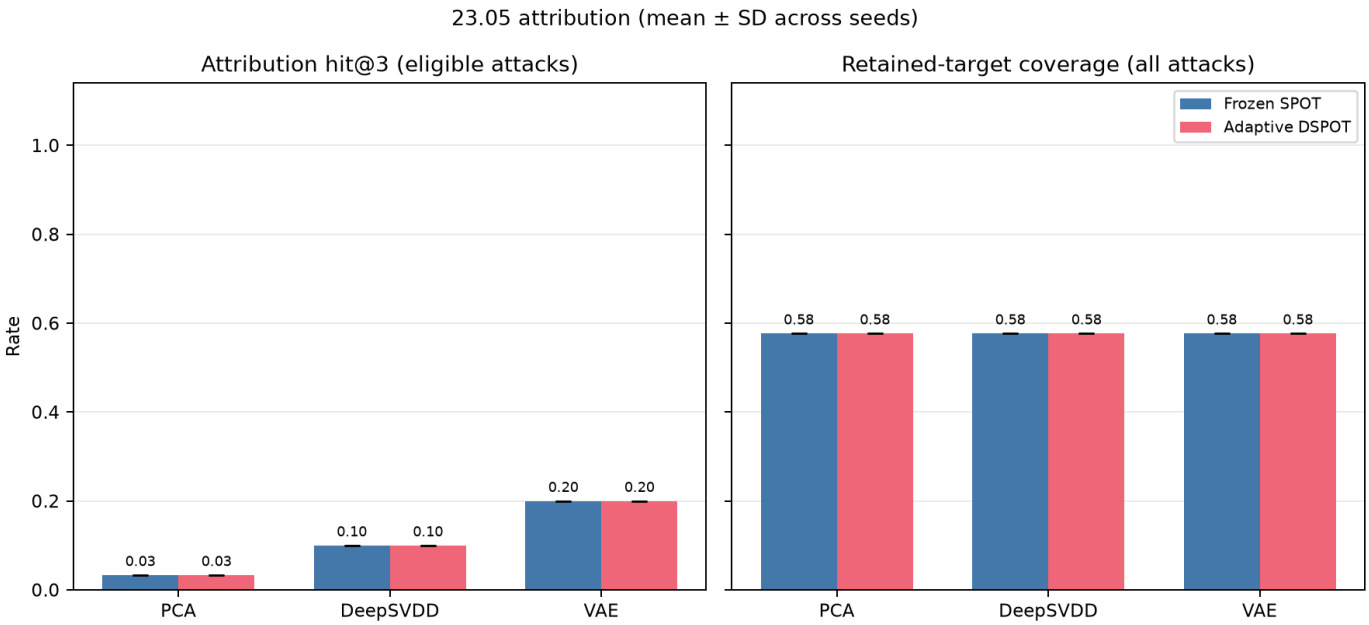
Attack	File	Duration (s)	Missed runs	PCA F	PCA A	DeepSVDD F	DeepSVDD A	VAE F	VAE A
A226	test2	119	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A227	test2	133	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A228	test2	156	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A229	test2	116	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A230	test2	155	3/18	3/3	3/3	3/3	3/3	3/3	0/3
A231	test2	96	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A232	test2	154	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A234	test2	530	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A235	test2	87	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A236	test2	120	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A237	test2	190	6/18	3/3	3/3	3/3	3/3	0/3	0/3
A238	test2	123	6/18	3/3	3/3	3/3	3/3	0/3	0/3



Per-attack 23.05 detection rates for every model and arm.

Attribution Results and Failures

Of the 52 unique 23.05 attacks, 30 retained a documented target in the 41-tag source-nonconstant modeling schema and 22 did not. The latter are excluded from hit-rate denominators, not counted as attribution failures.



Cross-domain attribution hit rate and retained-target coverage.

Detection versus target attribution

For attacks with a retained documented target, the contingency table crosses alarm-based detection with attribution hit@3. Each cell reports attack-seed rows / unique attacks, so persistent per-attack patterns are not hidden by pooling over seeds.

Model	Arm	Eligible rows	Not detected + hit	Not detected + no hit	Detected + hit	Detected + no hit
PCA	Frozen	90	0 / 0	0 / 0	3 / 1	87 / 29
PCA	Adaptive	90	0 / 0	0 / 0	3 / 1	87 / 29
DeepSVDD	Frozen	90	0 / 0	0 / 0	9 / 3	81 / 27
DeepSVDD	Adaptive	90	0 / 0	0 / 0	9 / 3	81 / 27
VAE	Frozen	90	18 / 6	54 / 18	0 / 0	18 / 6
VAE	Adaptive	90	18 / 6	57 / 19	0 / 0	15 / 5

VAE / frozen had the largest not detected + target hit cell: 18 attack-seed rows from 6 unique attacks (test1/A102, test1/A106, test2/A204, test2/A205, test2/A206, test2/A232). PCA / frozen had the largest detected + no target hit cell: 87 attack-seed rows from 29 unique attacks (test1/A102, test1/A103, test1/A104, test1/A105, test1/A106, test1/A109, test1/A110, test1/A111, test1/A112, test1/A113, test2/A202, test2/A203, test2/A204, test2/A205, test2/A206, test2/A207, test2/A208, test2/A214, test2/A215, test2/A216, test2/A218, test2/A219, test2/A225, test2/A226, test2/A227, test2/A228, test2/A230, test2/A231, test2/A233). Thus a documented target sensor can have a top-3 score response without the max-fused score crossing the EVT threshold; conversely, a fused alarm does not imply a correct target attribution. Detected-only delay statistics use a different attack subset: they omit target-responsive but undetected attacks, while detected attacks without a target hit remain eligible for delay summaries.

Per-attack attribution failures

The following table lists every eligible attack with at least one hit@3 miss. Cells give successful attributions / total seeds. Attribution and detection failures are distinct: an attack can be detected without a correct top-sensor attribution, and an attribution rank can be computed even when no alarm fires. Attacks with no retained documented target do not appear in this failure table by design.

Attack	File	Retained targets	Missed runs	PCA F	PCA A	DeepSVDD F	DeepSVDD A	VAE F	VAE A
A102	test1	P1_PIT01	12/18	0/3	0/3	0/3	0/3	3/3	3/3
A103	test1	P1_PCV01D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A104	test1	P1_PCV01D, P1_PIT01	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A105	test1	P1_PCV01D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A106	test1	P1_PIT01	12/18	0/3	0/3	0/3	0/3	3/3	3/3
A109	test1	P1_FT03	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A110	test1	P1_FCV03D	18/18	0/3	0/3	0/3	0/3	0/3	0/3

Attack	File	Retained targets	Missed runs	PCA F	PCA A	DeepSVDD F	DeepSVDD A	VAE F	VAE A
A111	test1	P1_FCV03D, P1_FT03	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A112	test1	P1_FCV03D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A113	test1	P1_FT03, P1_LIT01	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A202	test2	P1_LIT01	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A203	test2	P1_LCV01D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A204	test2	P1_LCV01D, P1_LIT01	12/18	0/3	0/3	0/3	0/3	3/3	3/3
A205	test2	P1_LCV01D	12/18	0/3	0/3	0/3	0/3	3/3	3/3
A206	test2	P1_PIT01	12/18	0/3	0/3	0/3	0/3	3/3	3/3
A207	test2	P1_LCV01D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A208	test2	P1_LCV01D, P1_LIT01	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A214	test2	P1_FCV01D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A215	test2	P1_FCV01D, P1_TIT01	12/18	0/3	0/3	3/3	3/3	0/3	0/3
A216	test2	P1_FCV01D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A218	test2	P1_PCV01D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A219	test2	P1_FCV03D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A225	test2	P1_LCV01D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A226	test2	P1_FCV03D, P1_PCV01D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A227	test2	P1_FT03	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A228	test2	P1_PCV01D, P1_PIT01, P2_SIT01	12/18	0/3	0/3	3/3	3/3	0/3	0/3
A230	test2	P1_FCV01D, P1_PIT01	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A231	test2	P3_LCP01D	18/18	0/3	0/3	0/3	0/3	0/3	0/3
A233	test2	P1_LCV01D, P1_LIT01	18/18	0/3	0/3	0/3	0/3	0/3	0/3

Statistical Checks (added)

These checks were run after the original analysis. They reuse the saved per-window and per-sensor scores; no model was retrained. Scripts and raw outputs are in `analysis/` and `results/`.

Is the event-detection rate above chance?

A circular shift rolls each model's alarm series within its stream, keeping the run structure and destroying only its alignment with the attacks (200 shifts per configuration).

Model	Domain	Arm	Observed	Null mean	p
pca	21.03	frozen	0.740	0.032	0.000
pca	21.03	adaptive	0.480	0.016	0.000
pca	23.05	frozen	1.000	1.000	1.000
pca	23.05	adaptive	1.000	1.000	1.000
svdd	21.03	frozen	0.787	0.055	0.000
svdd	21.03	adaptive	0.813	0.062	0.000
svdd	23.05	frozen	1.000	1.000	1.000
svdd	23.05	adaptive	1.000	1.000	1.000
vae	21.03	frozen	0.560	0.020	0.000
vae	21.03	adaptive	0.340	0.012	0.000
vae	23.05	frozen	0.154	0.020	0.000
vae	23.05	adaptive	0.115	0.009	0.000

On 23.05 the DeepSVDD and PCA event recall of 1.000 is exactly what a randomly time-shifted alarm series achieves (p = 1.000). It is alarm collapse, not detection.

PR-AUC with 95% block-bootstrap intervals

Model	Domain	PR-AUC	95% CI
pca	21.03	0.502	[0.404, 0.607]
pca	23.05	0.110	[0.037, 0.183]
svdd	21.03	0.486	[0.383, 0.598]
svdd	23.05	0.092	[0.028, 0.157]
vae	21.03	0.462	[0.372, 0.556]
vae	23.05	0.232	[0.068, 0.381]

No in-domain interval overlaps its cross-domain counterpart: the degradation is significant, not a shift in the mean.

Paired tests on event recall

Domain	Comparison	Delta recall	p
21.03	pca vs svdd	-0.047	0.015
21.03	pca vs vae	+0.180	0.000
21.03	svdd vs vae	+0.227	0.000
23.05	pca vs svdd	+0.000	1.000
23.05	pca vs vae	+0.846	0.000
23.05	svdd vs vae	+0.846	0.000

Fusion ablation: is max fusion the cause of the collapse?

Model	Fusion	PR-AUC	Recall	FA/day
pca	max	0.110	1.000	8297
pca	mean	0.128	1.000	8297
pca	median	0.121	1.000	8297
pca	top3mean	0.106	1.000	8196
svdd	max	0.092	1.000	8297
svdd	mean	0.136	1.000	8297
svdd	median	0.039	0.513	1629
svdd	top3mean	0.094	1.000	8297
vae	max	0.232	0.154	16
vae	mean	0.247	0.154	15
vae	median	0.108	0.032	8
vae	top3mean	0.243	0.154	12

The collapse survives every fusion rule: PCA and Deep SVDD alarm continuously under max, mean and top3-mean. Max fusion is not the cause; the representation shift is. Mean fusion does improve cross-domain PR-AUC modestly and is worth adopting.

In-domain calibration already misses its target

Model	Source split /day	21.03 test /day	Ratio to target
pca	9.7	30.1	3.0x
svdd	13.5	74.0	7.4x
vae	10.6	18.3	1.8x

The EVT layer hits 10 alarms/day on its own calibration split and then misses by 1.8-7.4x on a different period of the same plant version. The cross-version failure amplifies an existing in-domain gap.

Are the documented targets consistent with the process labels?

The documented target shares a process with the CSV per-process attack labels in **49 of 50** 21.03 attacks. The single exception is A220, where the documentation lists P2 while the CSV labels P3 and P4 — an internal inconsistency in the documentation, not a transcription error. 23.05 ships a single label column, so this check applies to 21.03 only.

Attribution: median rank and target type

Model	Domain	Median target rank (of 41; chance 21)
pca	21.03	8.0
pca	23.05	17.5
svdd	21.03	6.0
svdd	23.05	15.0
vae	21.03	7.0
vae	23.05	19.0

Target type (23.05)	Median rank
actuator	25.0
actuator+sensor	10.0
actuator+sensor+setpoint	8.5
actuator+setpoint	21.5
sensor+setpoint	13.0

PCA's Hit@3 of 0.033 on 23.05 reads as useless, but its median rank of 17.5 is still better than chance. Hit@3 alone overstates the failure.

Conclusions

- Across models and seeds, frozen event detection moved from 0.696 on 21.03 to 0.718 on 23.05 (loss -0.022); the adaptive arm reached 0.705 on 23.05 (signed gain -0.013).
- The highest raw frozen 23.05 event recall belonged to PCA and DeepSVDD (1.000 ± 0.000); the highest raw adaptive recall belonged to PCA and DeepSVDD (1.000 ± 0.000). Raw recall is not a stand-alone measure of useful detection.
- PCA and DeepSVDD had the largest signed mean DSPOT event-detection gain (+0.000; ratio-of-means recovery undefined). A positive gain is recovery, whereas a negative gain is an additional loss.
- VAE achieved the highest mean 23.05 PR-AUC (0.232 ± 0.000). Frozen and adaptive PR-AUC are identical because both arms threshold the same continuous scores.
- For VAE, frozen event recall moved from 0.560 on 21.03 to 0.154 on 23.05; DSPOT changed cross-domain recall to 0.115 (gain -0.038; recovery ROM -0.095) and false alarms from 16.41 to 5.47 per day.
- Observed 23.05 false alarms spanned 5.470–8296.588 per day across runs; the source-normal calibration target was 10 per day. Near-continuous-alarm failure screen (at least 99.9% event recall and at least 90% normal-window false alarms): PCA / frozen (mean recall 1.000, mean FA/day 8296.59, normal-window alarm fraction 96.0%); PCA / adaptive (mean recall 1.000, mean FA/day 8296.59, normal-window alarm fraction 96.0%); DeepSVDD / frozen (mean recall 1.000, mean FA/day 8296.59, normal-window alarm fraction 96.0%); DeepSVDD / adaptive (mean recall 1.000, mean FA/day 8296.59, normal-window alarm fraction 96.0%). These high-recall conditions are alarm-collapse failures, not evidence of useful detection or DSPOT recovery.

The central interpretation is decision-centric. **DSPOT adapts the EVT decision layer, not the encoder, PCA basis, or anomaly-score representation.** Thus a thresholded-metric gain is evidence of decision adaptation at the calibrated operating objective, not evidence of improved representation quality. The arm-invariant PR-AUC and ROC-AUC make that distinction explicit. False-alarm rates, event failures, and delay must therefore be read together rather than selecting a detector from PR-AUC alone.

Limitations

- The evidence concerns one HAI testbed, two plant versions, and three random seeds; standard deviations describe seed variability but are not confidence intervals or significance tests.
- Shared-tag intersection and source-nonconstant filtering change attribution eligibility. Target coverage must accompany hit rates, and missing documented targets cannot be interpreted as model errors.
- Attack documentation and retained targets are incomplete proxies for physical root cause; a top-k hit does not establish causal attribution.

- The evaluation is an offline chronological replay. Windows are causal, but computational latency and online control-loop safety are not measured.
- Source-normal EVT calibration controls an empirical source alarm rate, not a guaranteed 23.05 false-alarm budget. DSPOT may also adapt to attacks or nonstationarity without labels.
- Mean and median delays condition on detected attacks; comparisons are optimistic for conditions with many misses and should be read alongside event recall.

Reproducibility

From the repository root, with the package and locked dependencies installed, run:

```
hai-prepare --config configs/default.yaml
hai-train --config configs/default.yaml --model all
hai-evaluate --config configs/default.yaml --root results
hai-report --root results --reports reports --figures figures
```

Generated artifacts are `metrics_summary.csv`, `degradation_recovery.csv`, and `recovery_summary.csv` under the result root; six PNGs under the figure directory; and Markdown, PDF, and poster outputs under the report directory. The report generator validates the full three-seed result matrix and rejects smoke metadata.

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7. W.-S. Hwang, J.-H. Yun, J. Kim, and B.-G. Min. “Do You Know Existing Accuracy Metrics Overrate Time-Series Anomaly Detections?” *37th ACM/SIGAPP Symposium on Applied Computing (SAC)*, 2022. Introduces eTaPR; DOI: 10.1145/3477314.3507024.
8. S. Kim, K. Choi, H.-S. Choi, B. Lee, and S. Yoon. “Towards a Rigorous Evaluation of Time-Series Anomaly Detection.” *AAAI Conference on Artificial Intelligence*, 2022.